

Organizational-memory claims under original-world grouped validation

A V4.1 reconciliation of a committed simulation evidence package

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Manuscript status. No new simulation was run. V4.0 is preserved unchanged from Git commit 23b53aee4169deeda30aad2a9dba8587024f8d3d. V4.1 changes the statistical analysis and the authorized interpretation, not the underlying trajectories.

Abstract

Computational evidence can contain many observations but few independent experimental worlds. We re-examined a committed V4.0 package that reported decoding of a globally imposed cumulative-drive coordinate from component features after extensive material replacement. The audit used the original simulation seed as the outer validation group and did not run or replay any simulation. The V4.0 point estimate and interval were generated by resampling 12 drive histories with replacement and assigning duplicate draws new cross-validation identifiers. All 3,000 realized bootstrap replicates contained at least one duplicate history, allowing exact rows from an original world to occur in both training and testing.

Under leave-one-original-world-out validation, the deep-turnover cumulative-drive point estimate is $R^2=0.695$ across 36 records from three original worlds. The three held-out-world scores are 0.745, 0.414, and 0.925. A descriptive world-fold t interval is [0.051, 1.338], and the exact fixed-prediction world-block percentile range is [0.486, 0.886]. With only three worlds, neither range is treated as a reliable nominal 95% confidence interval. The prior claim that a lower bound clears the 0.50 qualification threshold is withdrawn. The deep-turnover order-contrast estimate is $R^2=-1.118$ and is not established.

Direct counts of 36/36 recorded surviving trajectories, zero recorded reassignment switches, and mean surviving original-material fraction 0.190 are reproduced. However, the committed summary artifact does not include component masks, association edges, alternatives, or fusion events, so event-level continuity cannot be independently reconstructed. Tracker sensitivity changes the deep cumulative-drive point only modestly relative to world-to-world variation. Because the drive was imposed globally, this supports global informational content under either component-selection rule, not local storage.

The scientifically reconciled conclusion is therefore narrow: a positive out-of-world point estimate for cumulative-drive information survives, while the V4.0 certification, fusion-free continuity claim, and local-storage interpretation do not. All claims and numerical changes are enumerated in a machine-readable ledger.

Keywords: grouped validation; simulation audit; information decoding; material turnover; leakage; reproducibility; uncertainty

1. Purpose and scope

V4.1 asks one question: **Do the published V4 claims and numerical values remain reproducible under strictly grouped, leakage-free statistical inference?**

The answer has two parts. First, the committed records reproduce the material fractions, survival counts, switch counts, and the V4.0 history-grouped point estimates. Second, the load-bearing uncertainty calculation does not survive when independence is defined at the level at which stochastic worlds were generated. The corrected analysis consequently narrows both the numerical claim and its interpretation.

This is an artifact reconciliation, not an extension of the simulation programme. It contains no unsealed prospective result. It makes no claim of identity, individual memory, reproduction, heredity, active reconstruction, or evolution.

2. Canonical evidence and preservation

The canonical V4.0 source is the Git object 23b53aee4169deeda30aad2a9dba8587024f8d3d, originally at branch `release/organizational-memory-v1`. The V4.0 release manifest lists 37 files. Every listed file was extracted from its canonical Git blob and verified against the manifest SHA-256 value. The preserved V4.0 directory contains the manuscript source and PDF, supplement source and PDF, claim ledger, statistical reaudit, release metadata, reproduction package, generated outputs, and the committed longitudinal raw artifact.

The primary longitudinal artifact contains 36 rows: 12 globally imposed drive histories observed in each of three simulation-seed worlds. The simulation seed is therefore the original-world identifier. Rows sharing a seed are not treated as independent outer-validation observations.

Secondary committed artifacts were used only for bounded reanalysis:

worlds and one fixed recipient body;

worlds, generated with largest-component reselection.

- a common-recipient transplant artifact with 36 records from three donor
- an in-place response artifact with 24 records from two original worlds;
- a historical body-baseline artifact with 48 records from four original

No missing event trajectory was reconstructed, and no simulator or physics engine was called.

3. Statistical reconciliation

3.1 Leakage mechanism in V4.0

V4.0 resampled the 12 history groups with replacement. Each draw was copied into a new bootstrap data set and assigned a new fold identifier. When one history was drawn more than once, identical rows from the same original worlds received different fold identifiers. A duplicate could then be included in training while its exact counterpart was included in testing.

The probability that 12 draws with replacement from 12 histories are all unique is $12!/12^{12} = 0.0000537$. Thus duplication is expected in 99.9946% of replicates. Under the fixed V4.0 bootstrap seed, all 3,000 realized replicates contained at least one duplicate. The old bootstrap distribution therefore does not represent leakage-free uncertainty for prediction in a new original world.

This finding does not imply that every V4.0 point estimate is numerically meaningless. It means that the resampled confidence interval and the certification derived from it are not valid for the stated generalization target.

3.2 Frozen V4.1 estimator

V4.1 uses ridge regression with penalty $\lambda=1$, matching the canonical model family. For each outer fold:

1. one complete original world is held out; 2. feature means, standard deviations, and constant-column decisions are estimated from the remaining worlds only; 3. the model is fit on the remaining worlds; 4. predictions are generated once for the held-out world; 5. no row from the held-out original world enters model fitting or preprocessing.

The primary score is pooled out-of-world R^2 . The mean of the normalized held-out-world losses gives the same displayed fold average. A stricter crossed sensitivity additionally excludes the matching drive history from the training worlds for each test row.

3.3 Uncertainty without bootstrap refitting

There are only three original worlds in the primary artifact. V4.1 therefore shows the three fold scores directly. It also reports:

- a descriptive Student-t interval over original-world fold scores; and

- exact ordered resampling of the three fixed out-of-world fold scores.

The second calculation has only $3^3=27$ ordered resamples. It does not retrain a cross-validation model after duplicating worlds. Neither calculation is presented as a reliable nominal 95% interval or used to certify the 0.50 threshold. The inferential limitation is the small number of original worlds, not the number of rows.

4. Results

4.1 The V4.0 bootstrap reproduces but is leakage-affected

The canonical reproduction package regenerates the reported V4.0 longitudinal point estimates, including deep cumulative-drive $R^2=0.8878$ and deep order-contrast $R^2=-0.2394$. It also regenerates the history-bootstrap interval [0.8366, 0.9581] for the cumulative coordinate. Code-path inspection and a deterministic audit of the 3,000 bootstrap samples show that every replicate contains duplicate histories relabelled as distinct folds. The numerical output is reproducible; its interpretation as leakage-free uncertainty across new worlds is not.

4.2 Original-world grouping lowers the deep headline and widens uncertainty

The cumulative-drive estimate remains positive at every recorded checkpoint under leave-one-original-world-out validation. It is 0.7382 initially, 0.9020 at moderate replacement, 0.8432 at the first deep checkpoint, and 0.6947 at the deep checkpoint. The corresponding crossed world-and-history exclusion estimates are 0.6021, 0.8842, 0.8041, and 0.6420.

At the deep checkpoint, the three held-out-world scores are:

- world 38502: 0.7454;
- world 38503: 0.4141;
- world 38504: 0.9246.

The descriptive t interval is [0.0513, 1.3381]. Exact resampling of the three fixed fold scores gives percentile endpoints [0.4859, 0.8858]. Because only three independent worlds are present, these ranges primarily display heterogeneity. Neither supports a lower bound above the 0.50 qualification threshold. The V4.0 CERTIFIED disposition is withdrawn.

The order-contrast estimate is 0.7631 initially but becomes -0.5102 at moderate replacement, -0.1315 at the first deep checkpoint, and -1.1183 at the deep checkpoint. Deep order information is not established.

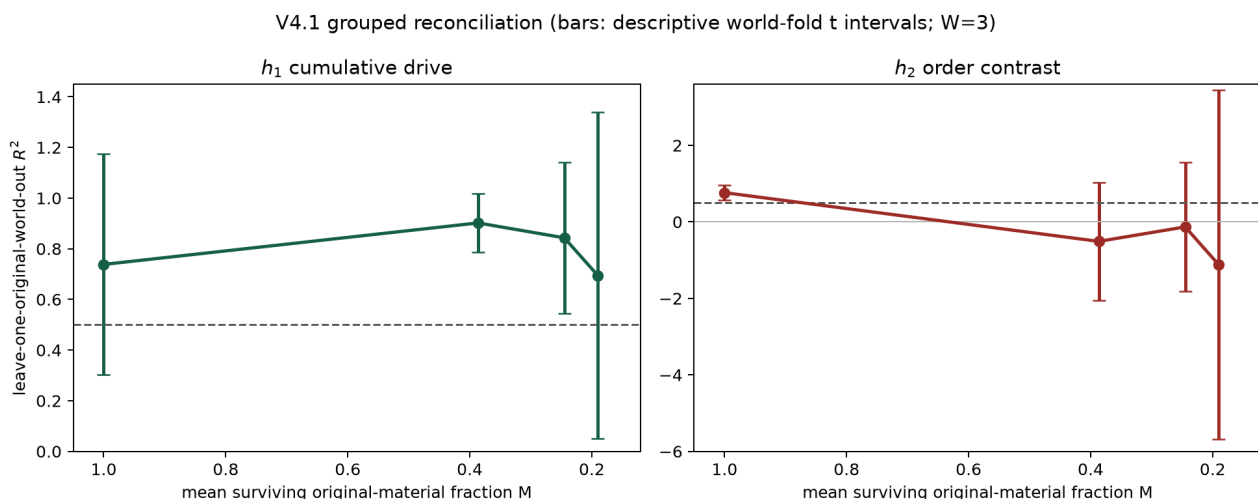


Figure 1. Original-world grouped longitudinal reconciliation. Points are pooled leave-one-original-world-out R^2 values from $n=36$ records, $W=3$ original worlds, and $H=12$ globally imposed histories at each checkpoint. Bars are descriptive t intervals over the three original-world fold scores, not nominal high-confidence intervals. The dashed line is the predeclared $R^2=0.50$ qualification threshold. Mean surviving original-material fractions are shown on the horizontal axis; the deep mean is $M=0.1902$.

4.3 Tracker continuity and fusion correction

The committed summary artifact directly reproduces 36/36 recorded surviving trajectories, zero lost records, zero recorded reassignment switches, mean deep $M=0.1902$, and 34/36 deep records at or below $M=0.25$.

The deep cumulative-drive estimate is 0.6947 with longitudinal tracking and 0.6706 with largest-component selection at each frame. Across the three worlds, the paired longitudinal-minus-largest differences are -0.0474, 0.0419, and 0.0777, for a mean difference of 0.0241. Tracker choice is therefore small relative to observed world variation for this summary decode.

This agreement does not verify event-level continuity. The raw summary contains lost and switch flags, but not the component masks, candidate association edges, gate terms, ambiguity alternatives, or merge/fusion events needed to audit a fusion decision. V4.1 retains the recorded counts and withdraws the stronger fusion-free interpretation.

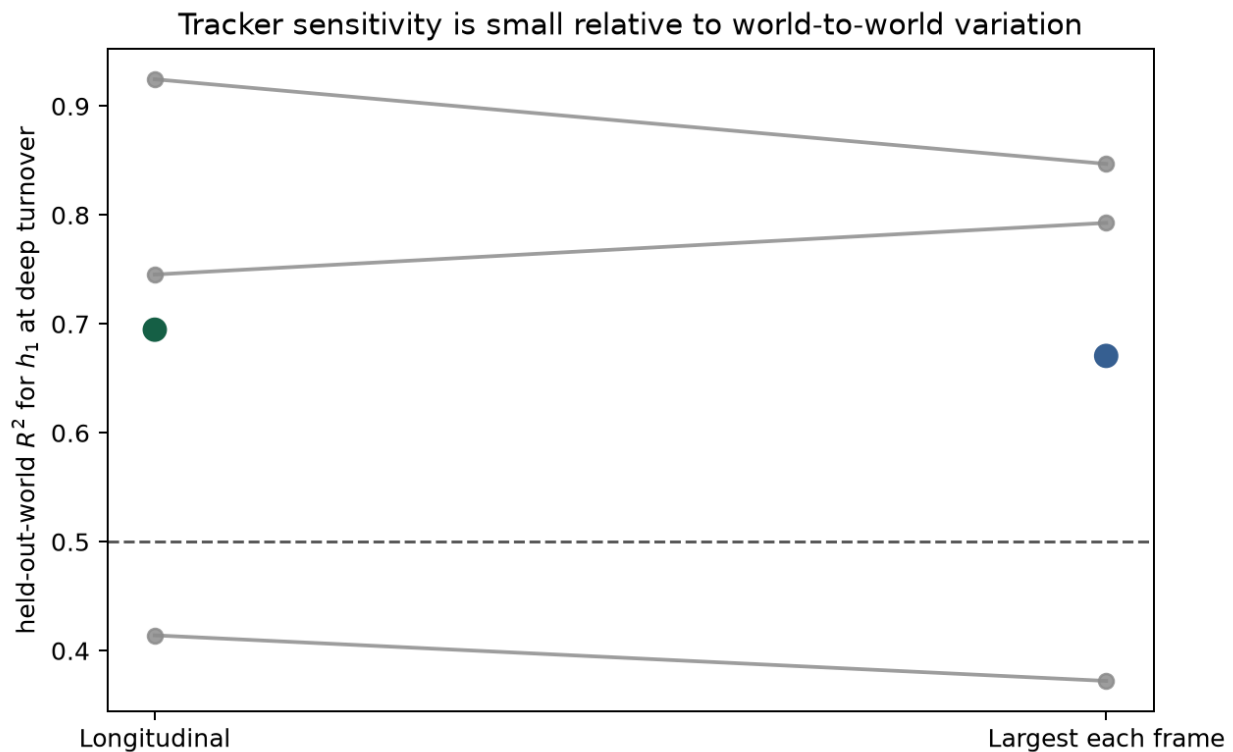


Figure 2. Tracker sensitivity at the deep checkpoint. Grey pairs are the held-out R^2 values for each of $W=3$ original worlds; colored points are pooled values from $n=36$ records and $H=12$ histories. Longitudinal tracking yields $R^2=0.6947$ and largest-component reselection yields $R^2=0.6706$. The plot supports a descriptive tracker-sensitivity statement only; absent event-level evidence, it does not certify fusion handling or identify a local storage site.

4.4 Secondary committed-artifact reanalyses

In the common-recipient transplant artifact, the fixed five-dimensional response vector gives cumulative-drive $R^2=0.6468$ under donor-world outer validation. Fold scores are 0.6568, 0.6972, and 0.5863 across $W=3$ donor worlds. The stricter crossed world-and-history estimate is 0.4807. The inert and erased responses are constant and score $R^2=0$ under world-held-out validation. These results support the bounded statement that globally written field values carry cumulative-drive information into one fixed recipient response and that readout removal eliminates that information. They do not establish generalization across recipient bodies.

The in-place response estimate is $R^2=0.7039$ but is based on $W=2$ worlds and is descriptive only.

In the historical $W=4$ body-baseline artifact, deep component-memory features score 0.7692 and size-plus-mass features score 0.4538. The paired mean difference is 0.3154, with a descriptive t interval $[-0.2394, 0.8701]$. The point favors component features, but the comparison is not separated. This artifact also used largest-component reselection and is not a direct local-versus-global access comparison.

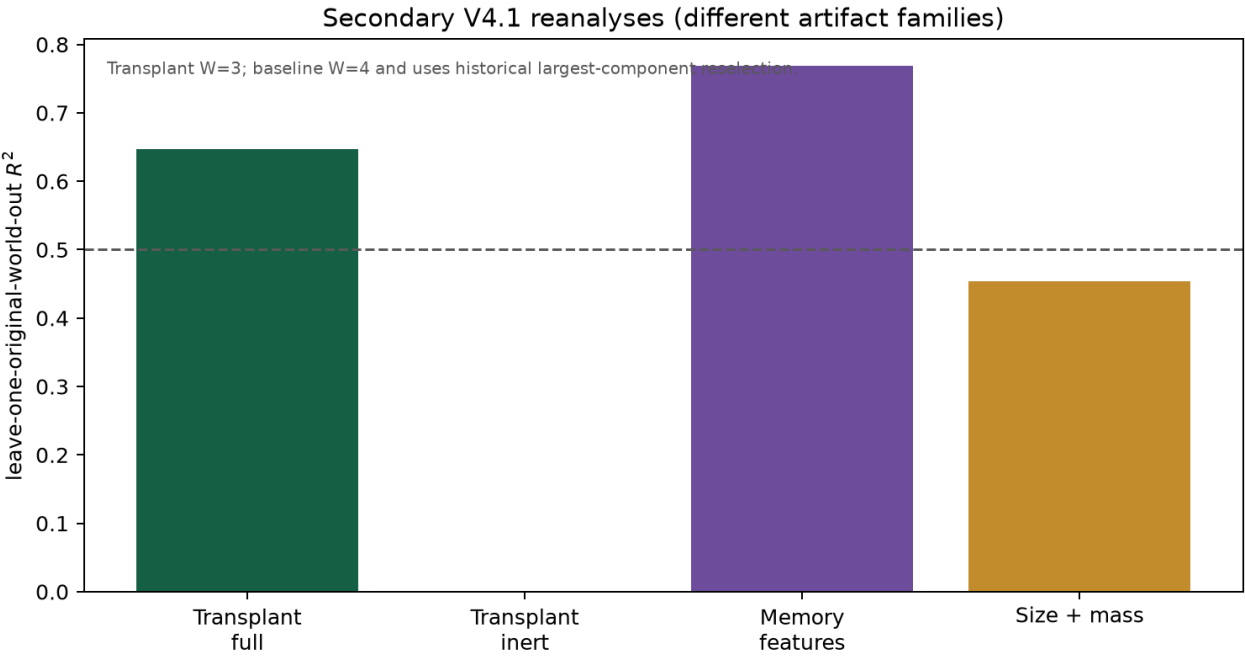


Figure 3. Secondary reanalyses from different committed artifact families. The transplant bars use $n=36$ records, $W=3$ donor worlds, $H=12$ histories, and one fixed recipient body. The deep body-baseline bars use $n=48$ records, $W=4$ original worlds, $H=12$ histories, and historical largest-component reselection. No uncertainty bars are combined because the bars do not share one experimental family. The dashed line marks $R^2=0.50$.

4.5 Global informational content is not a local-storage estimate

Every history in the primary design was imposed at whole-world level. A model trained on component features can therefore decode a world-level coordinate without demonstrating that the tracked component is the unique or local storage site. The V4 artifacts do not provide frozen access structures that compare the tracked component with neighbouring components, the surrounding environment, or whole-world features under the same outer folds.

The tracker comparison shows that two component-selection rules retain similar access to the globally imposed signal. It cannot distinguish whether the information is concentrated locally, redundantly distributed, or recoverable from broader world state. V4.1 consequently uses the term **global informational content** and withdraws local-storage wording.

5. Numerical and claim reconciliation

The following headline dispositions govern the manuscript, supplement, figures, machine-readable JSON, and cover letter.

Quantity	V4.0 canonical	V4.1 grouped result	Disposition
Deep cumulative-drive R^2	0.8878	0.6947	CORRECTED
Deep cumulative-drive interval	[0.8366, 0.9581]	descriptive t [0.0513, 1.3381]	WITHDRAWN
Deep threshold verdict	certified	not established	WITHDRAWN
Deep order-contrast R^2	-0.2394	-1.1183	CORRECTED
Track survival	36/36	36/36 recorded	UNCHANGED
Recorded switches	0	0	UNCHANGED
Mean deep M	0.1902	0.1902	UNCHANGED
Longitudinal / largest deep R^2	0.8878 / 0.9123	0.6947 / 0.6706	CORRECTED
Common-recipient transplant R^2	0.61	0.6468	CORRECTED
Inert / erased response R^2	-0.19 / -0.19	0 / 0	CORRECTED
Deep memory / size-plus-mass R^2	0.93 / 0.64	0.7692 / 0.4538	CORRECTED

Quantity	V4.0 canonical	V4.1 grouped result	Disposition
Fusion-free continuity	implicit	not reconstructable	WITHDRAWN
Local storage	stated	unresolved	WITHDRAWN

The complete reconciliation table and claim ledger are supplied as Markdown, CSV, and JSON. No narrative claim is authorized unless it matches that ledger.

6. Discussion

The reconciliation illustrates why the effective sample size in a simulation study is determined by the target of generalization. The primary artifact has 36 rows, but those rows arise from only three stochastic worlds. Histories are useful within-world interventions; they do not replace independent worlds when the claim concerns out-of-world prediction.

The positive cumulative-drive point is reasonably stable across two component-selection rules and remains positive under the crossed world-and-history exclusion sensitivity. That is meaningful evidence that the committed feature vectors contain global history-related information after substantial material replacement. It is not enough to estimate a reliable lower confidence bound, certify the 0.50 gate, prove event-level continuity, or locate storage.

The correction also separates reproducibility from validity. V4.0 is reproducible in the narrow computational sense: its package regenerates its numbers. The concern is methodological: duplicate bootstrap draws were treated as new folds, so the resulting interval does not answer the claimed out-of-world question. V4.1 preserves both facts rather than replacing the historical record.

Limitations

Only three original worlds support the primary headline, two support the in-place response result, and four support the historical body baseline. World-fold intervals are therefore descriptive. The summary artifact is insufficient for an event-level tracker audit. The common-recipient transplant does not estimate recipient-to-recipient generalization. The globally imposed histories do not identify a local storage site. Finally, the reanalysis is limited to committed feature vectors and cannot recover evidence that was never persisted.

7. Conclusion

Under strictly grouped, leakage-free outer validation, the V4.0 qualitative observation survives only in narrowed form: component features retain a positive out-of-world point estimate for a globally imposed cumulative-drive coordinate after extensive material replacement. The corrected deep estimate is $R^2=0.6947$ across three original worlds.

The V4.0 confidence interval and certification do not survive. Deep order information is not established. Recorded survival and switch counts reproduce, but event-level fusion handling cannot be reconstructed. The artifacts demonstrate global informational content, not local storage.

Final scientific verdict: MAJOR REPAIR STILL REQUIRED.

Data, code, and audit availability. The V4.0 blob-preserved package, V4.1 source, PDFs, analysis script, row-level out-of-fold predictions, numerical reconciliation table, errata ledger, claim ledger, headline-results JSON, cover letter draft, French summary, and visual QA report are contained in `paper/organizational-memory-v4-1-reconciliation/`.

The exact analysis and build commands are listed in `REPRODUCIBILITY.md`.